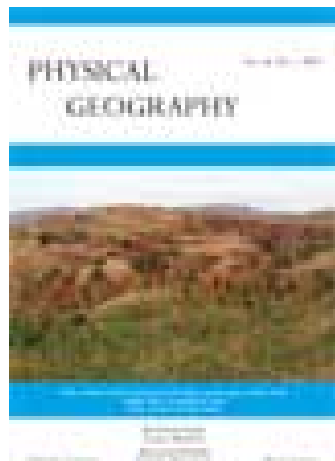




Physical Geography

Publication details, including instructions for authors and subscription information:

<http://www.tandfonline.com/loi/tphy20>

THE TOPOGRAPHIC RELATIVE MOISTURE INDEX: AN APPROACH TO SOIL-MOISTURE ASSESSMENT IN MOUNTAIN TERRAIN

Albert J. Parker ^a

^a University of Georgia , Athens

Published online: 15 May 2013.

To cite this article: Albert J. Parker (1982) THE TOPOGRAPHIC RELATIVE MOISTURE INDEX: AN APPROACH TO SOIL-MOISTURE ASSESSMENT IN MOUNTAIN TERRAIN, *Physical Geography*, 3:2, 160-168

To link to this article: <http://dx.doi.org/10.1080/02723646.1982.10642224>

PLEASE SCROLL DOWN FOR ARTICLE

Taylor & Francis makes every effort to ensure the accuracy of all the information (the "Content") contained in the publications on our platform. However, Taylor & Francis, our agents, and our licensors make no representations or warranties whatsoever as to the accuracy, completeness, or suitability for any purpose of the Content. Any opinions and views expressed in this publication are the opinions and views of the authors, and are not the views of or endorsed by Taylor & Francis. The accuracy of the Content should not be relied upon and should be independently verified with primary sources of information. Taylor and Francis shall not be liable for any losses, actions, claims, proceedings, demands, costs, expenses, damages, and other liabilities whatsoever or howsoever caused arising directly or indirectly in connection with, in relation to or arising out of the use of the Content.

This article may be used for research, teaching, and private study purposes. Any substantial or systematic reproduction, redistribution, reselling, loan, sub-licensing, systematic supply, or distribution in any form to anyone is expressly forbidden. Terms & Conditions of access and use can be found at <http://www.tandfonline.com/page/terms-and-conditions>

THE TOPOGRAPHIC RELATIVE MOISTURE INDEX: AN APPROACH TO SOIL-MOISTURE ASSESSMENT IN MOUNTAIN TERRAIN

Albert J. Parker
University of Georgia, Athens

Abstract: The Topographic Relative Moisture Index (TRMI), designed to indicate the relative soil moisture availability among sites in mountain terrain, is described. The TRMI is a summed scalar index of four slope parameters: topographic position, slope aspect, steepness, and slope configuration. A review of other methods of characterizing site moisture relations, including direct monitoring, water-balance climatology, site index, and other inferential topographic/edaphic indices, reveals that the simple, straightforward application of the TRMI is rapid, inexpensive, and effective in field research. The ability of the TRMI to portray significant patterns of plant response to environment is demonstrated from empirical studies at both the species-level (the ecology of Arizona cypress) and community-level (forest patterns in Yosemite National Park).

INTRODUCTION

Analyzing vegetation distribution patterns requires an accurate assessment of the spatial variability of environmental components that govern the distribution of plant species: water, light, heat, and nutrient availability (Walter, 1960; from Waring and Major, 1964). Because of its influence on photosynthetic processes, nutrient uptake, and the heat budget of plants (Barbour, Burk, and Pitts, 1980), water is often identified as a primary discriminating factor among terrestrial plant communities (Curtis, 1955; Hills, 1959; Marks and Harcombe, 1981).

Attempts to characterize site moisture conditions can be segregated into four approaches: direct monitoring (Barrs, 1968; Rawlins, 1976), water-balance climatology (Thornthwaite and Mather, 1955), site index determination (Avery, 1967; Husch, Miller, and Beers, 1972), and inferential techniques based on topographic or edaphic parameters (Whittaker, 1956; Loucks, 1962).

The use of direct monitoring of soil moisture includes both one-time measurement and continuous gauging of site conditions. Field use of single direct measures, such as gravimetric analysis (World Meteorological Organization, 1968), psychrometric techniques (Brown and McDonough, 1977), or tensiometric determination of soil or cell-sap water potential (Waring and Cleary, 1967; Cheung, Tyree, and Dainty, 1975), poses a serious methodological dilemma concerning the time and spacing of measurements. Soil moisture content and water potential fluctuate irregularly, because of precipitation and associated runoff and throughflow events, as well as diurnally and seasonally, because of temperature-induced changes in evapotranspirative flux from plants and soil. Hence, the vagaries of soil moisture additions and losses under field conditions seriously hinder the comparison of measurements taken at different times and places (Miller, 1977: 219-240). Continuous monitoring of moisture conditions on multiple sites is ideal (Barton, 1974); however, it is fiscally impractical under most field conditions, particularly given the large number of sample sites involved in most field biogeographic research.

Models based on water-balance climatology (Thornthwaite, 1948; Penman, 1948) relate temperature, radiation, wind, and precipitation to theoretical soil-water relations (deficit, surplus, utilization, and recharge). Such work has provided valuable insight into global and continental scale biogeographic patterns (Mather and Yoshioka, 1968; Meentemeyer, Box, and Thompson, 1982). Water-balance calculations, however, require climatic data from weather stations, or interpolation by triangulation among stations. The resolution of these data is not great enough to depict local site moisture conditions in complex terrain, where soil-water relations vary dramatically over short distances. In addition, these water-balance models do not account for the pronounced variability in soil moisture storage capacity encountered in the field.

For several decades, foresters have used the site index concept to assess site quality (Doolittle and Vimmerstedt, 1960). In site-index calculations, a measure of plant productivity, usually tree height, is employed to gauge site quality (of which soil moisture availability is a major component). By incorporating a direct measure of plant response, site index provides a functional assessment of site productivity, but it does not directly address the relationship between environmental factors and productivity. Furthermore, species productivity (and, hence, site index) varies with changes in macroclimate, necessitating the development of regional site-index curves that relate tree height to stand age for reference species. Because these site-index curves are commonly derived from plantation stock (even-aged, monocultural stands), site index is often poorly suited for use in native forests. Also, because site indices are developed for individual species, they can be compared to site indices for other species only by regression analysis. Hence, while site index provides a functional regional-scale assessment of forest productivity, it is of limited value in most biogeographic research because it is not easily adapted for use in mixed forests, it changes along climatic gradients, and its use is tied to the presence of specific tree species for which height-age curves have been previously constructed.

Inferential techniques based on topographic/edaphic conditions have been employed frequently by ecologists to characterize environmental moisture patterns while circumventing the difficulties of direct monitoring of moisture levels. Inferential methods are most often designed to provide an index along which individual sites may be ordered, based on measurable slope or soil characteristics. Sophistication ranges from simple ordering of topographic positions (e.g., draw/cove, north-facing slope, south-facing slope, ridgetop) to complex scalar indices which graphically combine a series of variables to yield a single measure of site conditions (e.g., the nomograms of Loucks (1962) or Wentworth (1981)). Noteworthy is the work of Whittaker (1956, 1960, 1967), which describes vegetation patterns in response to a "topographic complex-gradient" based on aspect and topographic position.

This paper describes a simple, straightforward technique called the Topographic Relative Moisture Index (TRMI; previously termed the Topographic Potential Moisture Index—Parker, 1980, 1982) which was developed for quantification of relative available moisture as dictated by topographic features. The TRMI represents an attempt, based solely on environmental parameters, to quantify the topographic complex-gradient among sites. My purpose in outlining this technique is not to argue for its acceptance, but rather to underscore the need for devising more direct approaches to field determination of such basic environmental resources as soil moisture, a development to which physical geographers can make important contributions.

THE TOPOGRAPHIC RELATIVE MOISTURE INDEX

Before the use of the TRMI is described, the constraints and scope of the work from which it has evolved must be specified. The index was developed for use in remote areas of highly dissected topography in middle latitudes of the northern hemisphere with local relief of several tens to several hundreds of meters. Hence, in its present form, the TRMI is not designed for use in flat or gently rolling landscapes. Its purpose is to allow integration of a number of slope characteristics into a one-dimensional scale of site moisture conditions. When combined with elevation, this topographic index allows an effective inferential assessment of relative site moisture levels, as dictated by landscape features. Development of the index was induced partially by a paucity of climate data; the density of even lower-order weather stations in mountainous areas is low, and the overwhelming majority of these are concentrated either in adjacent lowlands or on a few prominent peaks. In complex terrain, a mosaic of radiation, wind exposure, surface water flow, and drainage class variables which change dramatically over short distances combines to magnify the inadequacy of existing facilities for depicting local moisture patterns. A second constraint which encouraged development of this index was the need to maximize efficiency and economy in data collection with limited manpower and instrumentation capabilities.

Bearing these formulative qualifications in mind, the TRMI's logical underpinnings depend heavily upon hydrological inference. It is a scalar index determined by summing assigned values for the following set of slope parameters: topographic position, aspect, slope configuration, and steepness (Table 1). TRMI values may range from

Table 1. Determination of Topographic Relative Moisture Index Values

<i>Topographic position:</i>		<i>Slope steepness (degrees):</i>			
valley bottom	20	<3.0		10	
lower slope	15	3.0 to 5.9		9	
middle slope	10	6.0 to 8.9		8	
upper slope	5	*9.0 to 11.9		7	
ridge top	0	12.0 to 14.9		6	
		15.0 to 17.9		5	
		18.0 to 20.9		4	
<i>Slope configuration:</i>					
concave	10	21.0 to 23.9		3	
concave/straight	8	24.0 to 26.9		2	
straight	5	27.0 to 29.9		1	
convex/straight	2	≥30.0		0	
convex	0				
<i>Slope aspects (degrees azimuth):</i>					
19-26	20	81-89; 316-324	13	144-152; 253-261	6
27-35; 10-18	19	90-98; 307-315	12	153-161; 244-252	5
36-44; 1-9	18	99-107; 298-306	11	*162-170; 235-243	4
45-53; 352-360	17	108-116; 289-297	10	171-179; 226-234	3
54-62; 343-351	16	117-125; 280-288	9	180-188; 217-225	2
63-71; 334-342	15	126-134; 271-279	8	189-197; 208-216	1
72-80; 325-333	14	135-143; 262-270	7	198-207	0

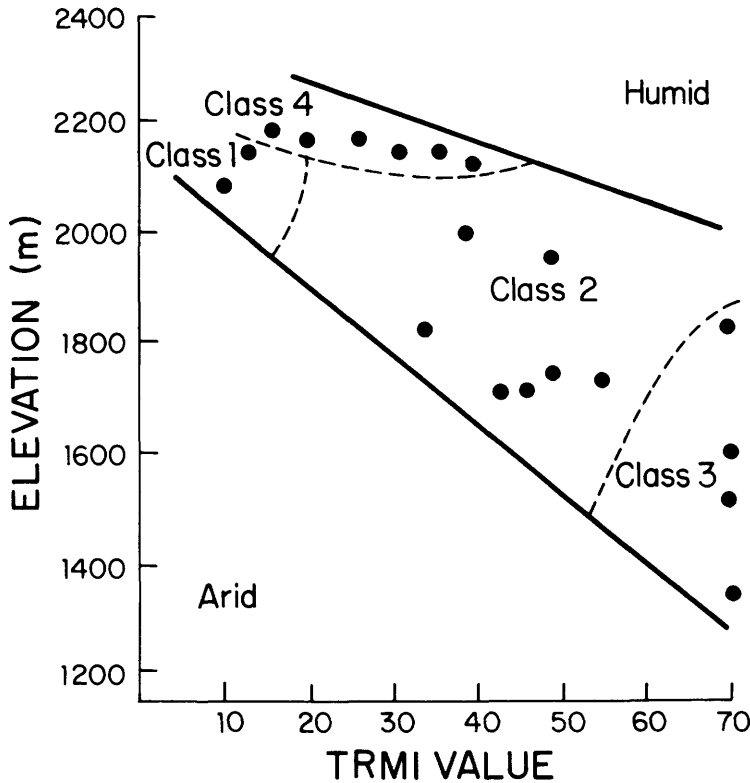


Fig. 1. Ecological distribution of Arizona cypress. A plot of stand TRMI values vs. elevation assists in identifying four geographic/ecological classes of Arizona cypress occurrence. The diagonal trend of the species habitat limit exhibits compensational interaction among elevation and topography.

0 (xeric sites) to 60 (mesic sites). For each stand, between 0 and 20 TRMI units are assigned for topographic position (ranging from 0 on exposed, well-drained ridge-tops to 20 in valley bottoms) and aspect (ranging from 0 on dry SSW-facing slopes of high solar radiation incidence to 20 on NNE-facing slopes); also, between 0 to 10 TRMI units are assigned for steepness (ranging from 0 on steep slopes exceeding 30° which foster rapid runoff, to 10 on slopes $>3^\circ$) and configuration (ranging from 0 on convex, or water dispersing slopes, to 10 on concave, or water collecting slopes). For example, a site located on a lower slope (15/20) which is concave downward and straight across (8/10), facing 240° azimuth (4/20) and sloping at 10° (7/10), would have a TRMI value of $15 + 8 + 4 + 7 = 34/60$ (see Table 1).

Although weighting of parameters is subjective, topographic position and slope aspect are weighted twice as heavily as the other factors in this index because both were judged to have strong influence on vegetation distribution, as inferred from field observation and previous literature (Cantlon, 1953; Whittaker, 1956; Hack and Goodlett, 1961).

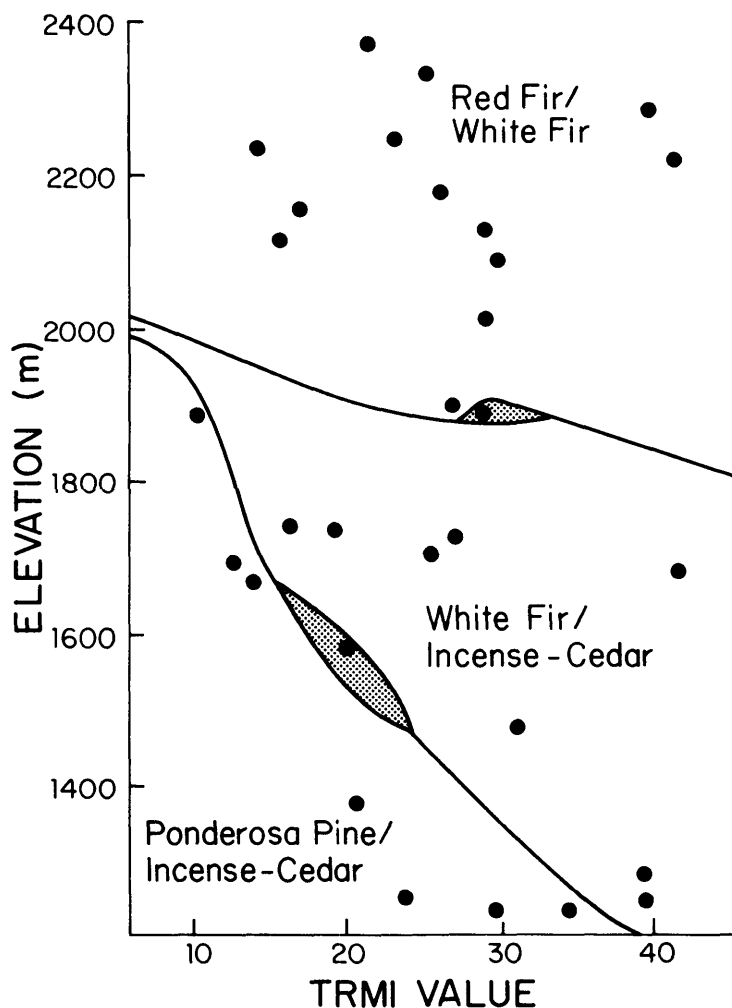


Fig. 2. Forest patterns in Yosemite National Park. A plot of stand TRMI values vs. elevation reveals that the dominant tree species range from ponderosa pine on xeric, lower elevation (1,200–2,000 m) sites, through white fir on mesic, lower elevation sites, to red fir at higher elevations (>2,000 m).

Applications of the Topographic Relative Moisture Index

The TRMI has been applied to both species-level and community-level biogeographic research. In work on the ecological distribution of Arizona cypress (*Cupressus arizonica*), a tree species of restricted occurrence in the American Southwest, the TRMI was found to be useful, when combined with elevation, in segregating geographic/ecologic classes of the species (Fig 1; Parker, 1980). Arizona cypress had been generally considered to be a riparian species in subhumid to semiarid canyons of uplands in southeastern Arizona. Although Arizona cypress does occur in these settings (Class

3), it may also be found in a variety of other topographic situations, including south-facing ridgetops at higher elevations. The diagonal trend of the lines through Figure 1, which delimit the species' ecological range, portrays compensational influences, with Arizona cypress being restricted to riparian settings and sheltered slopes at lower elevations, but occupying progressively drier sites (smaller TRMI values) with increasing elevation. Riparian settings confound the use of the TRMI, since the index assumes no direct groundwater-table influence on the site. Hence, in this case it was deemed appropriate to assign maximum values of 60 to riparian sites in lieu of considering individual slope features.

An examination of the distribution of forest types in the Yosemite National Park (Parker, 1982) further illustrates the utility of the TRMI. The arrangement of stands which results from plotting TRMI value against elevation clearly expresses the major site relations among three different forest types (Fig. 2). Ponderosa pine-dominated (*Pinus ponderosa*) forests occupy most slope positions below 1,500 m, but are progressively confined to steep, south- and west-facing upper slopes and ridgetops (smaller TRMI values) at higher elevations. Likewise, white fir-dominated (*Abies concolor*) forests are common on most settings between 1,500 and 2,000 m, but are limited to north-facing sites (higher TRMI values) at lower elevations. Red fir-dominated (*Abies magnifica*) forests become increasingly common above 2,000 m.

These two case studies demonstrate the effectiveness of the TRMI for examining the complex interactions between topography and elevation, and illustrate the TRMI's appropriateness for use in species and community level ecological research.

DISCUSSION

Most inferential methods of site-moisture assessment consider both topographic position and aspect/exposure as key variables in ordering stands along a topographic complex-gradient (Whittaker, 1956, 1960). Some workers have also included steepness as a relevant slope parameter in assessing site conditions (Wentworth, 1981), but the hydrologically important configuration parameter has been ignored (Hack and Goodlett, 1961). The TRMI is the only topographically based index which incorporates all four of these slope features. Wentworth's (1981) nomogrammatic technique, which fuses elevation with solar-irradiance (calculated from slope aspect and steepness, and modified by topographic position) to yield a single scalar-value of site energy levels, uses all but the configuration parameter, although aspect and elevation are preferentially weighted in his nomogram. In formulating the TRMI, elevation was excluded so as to characterize only the influence of local topography on site moisture.

One of the chief difficulties of Whittaker's (1956, 1960) topographic indices is that they rely on biological information for their infrastructure. In earlier works (Whittaker, 1956), position of a site along the topographic complex-gradient was determined by a weighted averaging of the presence of selected indicator plant species (Ellenberg, 1948) (i.e., species which occur with great "fidelity" in one of a graded sequence of sites). In later works (Whittaker, 1960; Whittaker and Niering, 1965), reliance on individual species as indicators was replaced by comparing a site's species composition with a composite sample of several sites from the mesic and xeric extremes of the data set. These methods pose several difficulties. First, in the earlier

work, ordering the gradient by indicator species requires prior knowledge of the ecological behavior of the species selected to act as indicators, thus limiting application of the technique to regions in which the investigator has intimate knowledge of plant ecological relations. Second, in either approach, a preconceived mental construct of the moisture relations among topographic settings is required to create and arrange the categories into which indicator species are assigned or from which the composite extremes are drawn. It is more logical to develop an index directly upon these preconceived mental constructs (as is the case for the TRMI), rather than filtering the gradient through the plant cover. This is especially true when the final use of the index is to examine plant species or vegetation distributions, since use of indicator species or composite extremes engenders circularity. One should either base topographic indices solely on nonbiological grounds and then proceed to interpret vegetation patterns from these (Wentworth, 1981; Parker, 1982), or use ordination techniques (Hill and Gauch, 1980) to construct a phytosociological index along which environmental variables can be correlated (Marks and Harcombe, 1981).

Initially, soil depth was used as a parameter in the TRMI, in order to incorporate root zone/storage capacity for sites; in fact, the study of Arizona cypress includes a 0 to 10 scale for increasing soil depth. Soil depth subsequently was removed from the index, in order to restrict the TRMI to slope parameters. Incorporation of soil depth, soil texture, and bulk density (cf. Waring and Major, 1964) into the TRMI, however, would expand the versatility and completeness of the index, without greatly increasing execution time in the field, and is being considered for future work.

Several problems and restrictions in applying the TRMI must be noted. First, while the weighting of slope parameters presented here seems justifiable from field experience and the ecological literature, arguments for different weighting schemes can be made. In fact, appropriate parameter weighting is likely to vary geographically, because of changes in average cloudiness and latitudinal influence on sun angle. A second problem with the index is intercorrelation or interactive influences among the slope parameters. For example, slope form commonly produces a correlation between topographic position and configuration, with upper slopes being convex and lower slopes being concave. Also, the importance of aspect in controlling solar radiation declines as slope steepness is reduced. Furthermore, use of the TRMI is restricted to situations in which variation in other factors is controlled. The index must be applied independently on each substrate type in regions where soils formed from different initial materials induce critical differences in soil moisture or nutrient availability (Whittaker, 1960; Wentworth, 1981). Finally, the index may fail to explain adequately the vegetation patterns in regions where disturbance and associated disparities in the stage of vegetation recovery superimpose temporal variability on the spatial pattern being depicted by the index.

In spite of these shortcomings, the TRMI provides valuable insights into the topographic control of vegetation patterns, as seen from the examinations of forest patterns in Yosemite and the ecological distribution of Arizona cypress. Although not quantitatively elegant, the TRMI embodies a simplicity which confers several benefits. Ease of use is one benefit. The requisite slope parameters can be measured rapidly and require only a compass/inclinometer for measurement. A second benefit is that, because the index is founded on well-established hydrological inference, it allows

direct interpretation of relative moisture availability without recourse to biological surrogates. A third benefit of this index is its sensitivity to microscale variation in topographic parameters, allowing comparisons of moisture conditions on a myriad of adjacent slope positions. Consequently, in addition to its applications in field biogeographic research, the speed and effectiveness of the TRMI in assessing site character is well suited to the preparation of environmental impact statements or the development of a site inventory on which resource management policies can be based.

Much work remains in the development of field techniques that quantify the spatial pattern of environmental resources. In developing these quantitative methods, however, the need to preserve simplicity in design and directness of application under field conditions must be recognized in order to facilitate the execution of the basic empirical work upon which theories must be founded.

BIBLIOGRAPHY

- Avery, T. E. (1967) *Forest Measurements*. New York: McGraw-Hill.
- Barbour, M. G., Burk, J. H., and Pitts, W. D. (1980) *Terrestrial Plant Ecology*. Menlo Park, CA: Benjamin/Cummings Publ. Co.
- Barrs, H. D. (1968) Determination of water deficits in plant tissues. In T. T. Kozlowski, ed., *Water Deficits and Plant Growth, Vol. I*. New York: Academic Press.
- Barton, M. (1974) New concepts in snow surveying to meet expanded needs. In *Interdisciplinary Symposium on Advanced Concepts and Techniques in the Study of Snow and Ice Resources (Proceedings)*. Washington, DC: National Academy of Sciences.
- Brown, R. W. and McDonough, W. T. (1977) Thermocouple psychrometer for *in situ* leaf water potential determinations. *Plant and Soil*, Vol. 48, 5-10.
- Cantlon, J. (1953) Vegetation and microclimates on north and south slopes of Cucke-tunk Mountain, New Jersey. *Ecol. Monogr.*, Vol. 23, 477-496.
- Cheung, Y. N. S., Tyree, M. T., and Dainty, J. (1975) Water relations parameters on single leaves obtained in a pressure bomb and some ecological interpretations. *Can. J. Bot.*, Vol. 53, 1342-1346.
- Curtis, J. T. (1955) A prairie continuum in Wisconsin. *Ecology*, Vol. 36, 558-566.
- Doolittle, W. T. and Vimmerstedt, J. P. (1960) *Site Index Curves for Natural Stands of White Pine in the Southern Appalachians*. U.S. For. Serv., Southeastern Forest Experimental Station Research Note 141.
- Ellenberg, H. (1948) Unkrautgesellschaften als Mass für den Säuregrad, die Verdichtung und andere Eigenschaften des Ackerbodens. *Bericht über Landtechnik*, Vol. 4, 130-146.
- Hack, J. and Goodlett, J. (1961) *Geomorphology and Forest Ecology of a Mountain Region in the Central Appalachians*. U.S. Geol. Surv. Prof. Paper 347, 66 pp.
- Hill, M. O. and Gauch, H. G., Jr. (1980) Detrended correspondence analysis: an improved ordination technique. *Vegetatio*, Vol. 42, 47-58.
- Hills, G. A. (1959) *A Ready Reference to the Description of the Land of Ontario and its Productivity*. Ontario Dept. Lands and Forests, 142 pp.
- Husch, B., Miller, C. I., and Beers, T. W. (1972) *Forest Mensuration* (2nd ed.). New York: Ronald Press.
- Loucks, O. L. (1962) Ordinating forest communities by means of environmental scalars and phytosociological indices. *Ecol. Monogr.*, Vol. 32, 137-166.

- Marks, P. L. and Harcombe, P. A. (1981) Forest vegetation of the Big Thicket, south-east Texas. *Ecol. Monogr.*, Vol. 51, 287-305.
- Mather, J. R. and Yoshioka, G. A. (1968) The role of climate in the distribution of vegetation. *Annals Assoc. Amer. Geogr.*, Vol. 58, 29-41.
- Meentemeyer, V., Box, E. O., and Thompson, R. (1982) World patterns and amounts of terrestrial plant litter production. *Bioscience*, Vol. 32, 125-128.
- Miller, D. H. (1977) *Water at the Surface of the Earth: An Introduction to Ecosystem Hydrodynamics*. New York: Academic Press.
- Parker, A. J. (1980) Site preferences and community characteristics of *Cupressus arizonica* Greene (Cupressaceae) in southeastern Arizona. *Southwestern Naturalist*, Vol. 25, 9-22.
- _____ (1982) Environmental and compositional ordination of conifer forests in Yosemite National Park, California. *Madrono*, Vol. 29, 109-118.
- Penman, H. L. (1948) Natural evaporation from open water, bare soil, and grass. *Proc. Roy. Soc. London (A)*, Vol. 193, 120-145.
- Rawlins, S. L. (1976) Measurement of water content and the state of water in soils. In T. T. Kozlowski (ed.), *Water Deficits and Plant Growth, Vol. IV*. New York: Academic Press.
- Thorntwaite, C. W. (1948) An approach toward a rational classification of climate. *Geographic Review*, Vol. 38, 55-94.
- _____ and Mather, J. R. (1955) *The water balance*. Drexel Inst. Techn., Lab. Climatology, Vol. 8, 1-104.
- Walter, H. (1960) *Standortslehre. Einführung in die Phytologie. Bd III. Grundlagen der Pflanzenverbreitung. Teil 1* (2nd ed.). Stuttgart: Ulmer, 566 pp.
- Waring, R. H. and Cleary, B. D. (1967) Plant moisture stress: evaluation by pressure bomb. *Science*, Vol. 155, 1248-1254.
- _____ and Major, J. (1964) Some vegetation aspects of the California coast redwood region in relation to gradients of moisture, nutrients, light, and temperature. *Ecol. Monogr.*, Vol. 34, 167-215.
- Wentworth, T. R. (1981) Vegetation on limestone and granite in the Mule Mountains, Arizona. *Ecology*, Vol. 62, 469-482.
- Whittaker, R. H. (1956) Vegetation of the Great Smoky Mountains. *Ecol. Monogr.*, Vol. 26, 1-80.
- _____ (1960) Vegetation of the Siskiyou Mountains, Oregon and California. *Ecol. Monogr.*, Vol. 30, 279-338.
- _____ (1967) Gradient analysis of vegetation. *Biological Review*, Vol. 42, 207-264.
- _____ and Niering, W. A. (1965) Vegetation of the Santa Catalina Mountains, Arizona: a gradient analysis of the south slope. *Ecology*, Vol. 46, 429-452.
- World Meteorological Organization (1968) *Practical Soil Moisture Problems in Agriculture*. Tech. Note 97, 69 pp.